ModelIt!

Student Activity Pack

Every Drop You Drink Is Recycled Dinosaur Water

The Never-Ending Journey of Water on Earth

Grade 7 | NGSS MS-ESS2-4 | Energy and Matter

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Where does the water you drink come from originally?

2 What happens to water in a puddle on a hot day?

3 Draw what you think happens to water as it moves through Earth's systems.

4 Could the water you drink today have once been in a dinosaur? Why or why not?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Solar Energy | Ground Temperature | Evaporation Rate | Precipitation Rate

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Summer Heat Wave

Maximum solar energy and ground temperature

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Winter Freeze

Ground Temperature near freezing, reduced solar energy

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Normal Conditions

Moderate solar energy and temperature

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Wind Patterns — Wind carries water vapor from oceans to land, determining where precipitation falls

Plant Transpiration — Plants release water vapor through their leaves, adding significant moisture to the atmosphere

Human Water Use — Humans withdraw water from rivers, lakes, and aquifers, affecting how much is available for natural cycling

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why doesn't Earth ever run out of water if we use billions of gallons every day? Use evidence from your simulation to support your answers.

1. **Why is the water cycle called a CYCLE and not a one-way process?**

2. **How does cutting down forests affect the water cycle in that region?**

3. **Why do some places get floods while nearby places have droughts at the same time?**

4. **What would happen to the water cycle if the Sun suddenly produced less energy?**

5. **How does understanding the water cycle help us manage water resources?**

8 STEM Challenge: Design a Water Purification System

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YOUR ENGINEERING MISSION

Engineer a portable water purification system using your understanding of the water cycle — specifically evaporation and condensation — to make dirty water drinkable.

THE SCENARIO

After a natural disaster, clean water supplies are contaminated. Your engineering team must design a solar-powered water purification system using water cycle principles to provide clean drinking water.

Engineering Design Thinking

1. HOW WILL YOU USE SOLAR ENERGY TO EVAPORATE CONTAMINATED WATER?
2. HOW WILL YOU CAPTURE AND CONDENSE THE WATER VAPOR?
3. HOW WILL YOU COLLECT THE PURIFIED WATER?
4. HOW WILL YOU TEST WHETHER YOUR PURIFIED WATER IS ACTUALLY CLEAN?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Hydrologists study the movement, distribution, and quality of water on Earth. They work for government agencies, environmental firms, and water utilities, earning $65,000–$110,000/year.